

Weighted Importance Sampling

This note derives equation (5.8) in Sutton and Barto's book. Instead W_k in the book, I use P_t as the importance sampling weight.

The agent is learning a target policy π from a behavior policy b . It has performed a number of episodes of training using b . It encounters a specific states many times during these behavior episodes. It lines up these behavior episodes one after the other as if they happen in continuous time. Specifically if episode i ends at time T_i , the assumption is that episode $i+1$ starts at time T_i+1 and ends at T_{i+1} . The last episode N ends at time T_N .



Using Weighted Importance Sampling, we can write (in general)

$$v_{\pi}(s) = \frac{\sum_{t \in \mathcal{U}(s)} P_t G_{t,b}}{\sum_{t \in \mathcal{U}(s)} P_t} \quad (1)$$

where $\mathcal{U}(s)$ is all the time points at which state s occurs

in behavior b episodes.

Let $G_{1,b}, G_{2,b}, \dots, G_{n-1,b}$ be a sequence of Returns, all starting at state s and each with weight p_k . Let us number the occurrences of state s in the behavior episodes as $1, 2, \dots, n$.

We can write modifying (1) a bit as

$$v_{\pi,n} = \frac{\sum_{k=1}^{n-1} p_{t_k,k} G_{t_k,b}}{\sum_{k=1}^{n-1} p_{t_k,k}} \quad \dots (2)$$

Dropping π, b, t and k subscripts to make the formula look simpler, we can write

$$v_n = \frac{\sum_{k=1}^{n-1} p_k G_k}{\sum_{k=1}^{n-1} p_k} \quad \dots (3)$$

We can also write by changing the subscript in (3) $n+1$,

$$v_{n+1} = \frac{\sum_{k=1}^n p_k G_k}{\sum_{k=1}^n p_k} \quad \dots (4)$$

Subtracting (3) from (4), we get

$$v_{n+1} - v_n = \frac{\sum_{k=1}^n p_k G_k}{\sum_{k=1}^n p_k} - \frac{\sum_{k=1}^{n-1} p_k G_k}{\sum_{k=1}^{n-1} p_k}$$

$$= \frac{\left(\sum_{k=1}^n p_k G_k \right) * \left(\sum_{k=1}^{n-1} p_k \right) - \left(\sum_{k=1}^n p_k \right) * \left(\sum_{k=1}^{n-1} p_k G_k \right)}{\left(\sum_{k=1}^n p_k \right) * \left(\sum_{k=1}^{n-1} p_k \right)}$$

$$= \frac{\left(\sum_{k=1}^{n-1} p_k G_k + p_n G_n \right) \left(\sum_{k=1}^{n-1} p_k \right) - \left(\sum_{k=1}^{n-1} p_k + p_n \right) \left(\sum_{k=1}^{n-1} p_k G_k \right)}{\left(\sum_{k=1}^n p_k \right) \left(\sum_{k=1}^{n-1} p_k \right)}$$

$$= \frac{p_n G_n \sum_{k=1}^{n-1} p_k - p_n \sum_{k=1}^{n-1} p_k G_k}{\left(\sum_{k=1}^n p_k \right) \left(\sum_{k=1}^{n-1} p_k \right)}$$

$$= \frac{p_n}{\left(\sum_{k=1}^n p_k \right)} \left[\frac{G_n \sum_{k=1}^{n-1} p_k}{\sum_{k=1}^{n-1} p_k} - \frac{\sum_{k=1}^{n-1} p_k G_k}{\sum_{k=1}^{n-1} p_k} \right]$$

$$= \frac{p_n}{\left(\sum_{k=1}^n p_k \right)} \left[G_n - \frac{\sum_{k=1}^{n-1} p_k G_k}{\sum_{k=1}^{n-1} p_k} \right]$$

$$= \frac{p_n}{C_n} [G_n - v_n] \quad \text{where } C_n = \sum_{k=1}^n p_n$$

This gives us

$$v_{n+1}(s) = v_n(s) + \frac{p_n}{C_n} [G_n - v_n(s)] \quad \dots \quad (5)$$

Formula (5) has been derived for updating v values in π incrementally from the n th occurrence of state s to the $n+1$ th occurrence of state s in the episodes of training using behavior policy b .

A similar formula can be obtained for updating $q(x, a)$ values (in π) incrementally from $\text{Return}(s, a)$ values in behavior b . The formula is

$$q_{n+1}(s, a) = q_n(s, a) + \frac{p_n}{C_n} [G_n(s, a) - q_n(s, a)] \quad \dots \quad (6)$$

Equation (6) is used by Barto and Sutton in the "Off-Policy MC Prediction" and "Off-Policy MC Control" algorithms toward the end of Chapter 6, page numbers 110 and 111, respectively.